BA2026 CAPSTONE PRESENTATION

AI-Enabled Business Impact Deck: KazMunayGas (KMG) Transformation Strategy

A practical AI adoption proposal for KMG, informed by Shell

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KMG should use AI to improve maintenance and field operations, but not for fuel price setting

CORE BOARD RECOMMENDATION

Focus AI investment on three areas, and keep Shell's pricing model out.

Predictive maintenance, ABAI reservoir management and TUMAR safety monitoring. Shell's approach to flexible fuel pricing and downstream asset sales should not be transferred directly to KMG.

WHY THIS FITS KMG

KMG has commercial targets and national responsibilities. Samruk-Kazyna holds 67.42% of the company, and KMG plays a major role in domestic fuel supply and regional employment.

1

PRIORITY 1

Refinery Predictive Maintenance

- Scale vibration and thermodynamic machine-learning sensors across Atyrau (ANPZ) and Pavlodar (PNHZ) refineries.
- Target: −15% downtime, below 102 h/year.

−15%

unplanned downtime

2

PRIORITY 2

ABAI Subsoil Platform Expansion

- Expand the ABAI digital twin platform across 1,660 waterflooding injection cells.
- Optimize artificial-lift power OPEX.

8–12%

OPEX reduction target

3

PRIORITY 3

Human-in-the-Loop Governance

- Introduce a four-category Human–AI task model.
- A qualified engineer reviews and approves critical equipment alerts.

4

task categories, engineer sign-off

KMG’s priorities include improving efficiency as well as maintaining production

REVENUE 2025

9,371 bn KZT

$17.98 billion

EBITDA 2025

2,393 bn KZT

$4.59 billion

NET DEBT / EBITDA

0.16x

Net debt ÷ EBITDA

NET DEBT

375 bn KZT

$720 million

CRUDE PRODUCTION

26.21 Mt

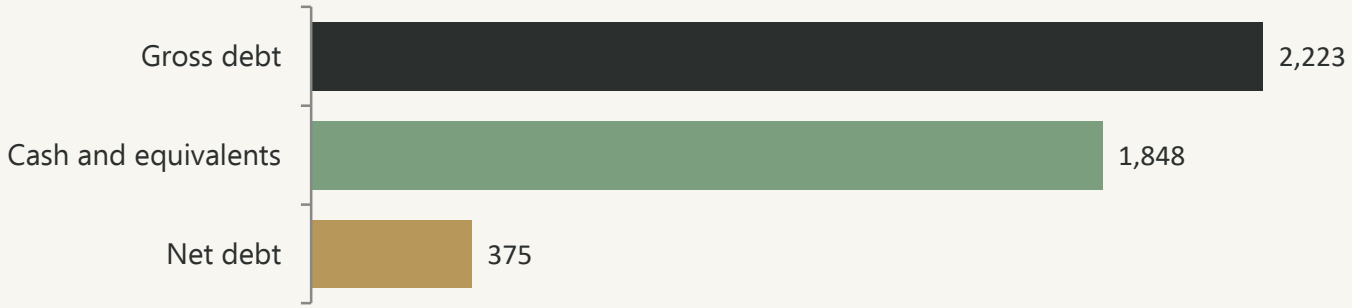
26% of Kazakhstan crude

TRANSPORTED VOLUME

83 Mt

55% of transported volume

Debt position, bn KZT (gross debt − cash = net debt)



WEEK 1: INITIAL VIEW

- Traditional crude exporter, vulnerable to oil price drops.
- Shell’s digital roadmap could be copied directly.
- Expected rapid refinery divestment and dynamic fuel pricing.
- Oversimplified Kazakhstan’s regulation and state mandates.

WHAT THE LATER ANALYSIS SHOWS

- KMG is a state-linked national company and a major domestic fuel supplier.
- Net Debt / EBITDA is 0.16x, with 1,848 bn KZT in cash and cash equivalents. This supports testing predictive maintenance at ANPZ and further use of ABAI.
- Order No. 171 price caps legally prohibit dynamic retail surge pricing.

KMG must balance commercial performance with domestic energy supply

1 OWNERSHIP AND SOVEREIGN MANDATE

67.42%

Samruk-Kazyna JSC

Requires national socio-economic stability and regional employment.

20.00%

Ministry of Finance

Fiscal dividend revenue and budget contribution.

9.58%

National Bank of Kazakhstan

National welfare fund asset preservation.

BUSINESS IMPLICATION

KMG cannot optimize margins at the expense of national fuel supply.

2 EXTERNAL MACRO FORCES (PESTEL)

P/L

Political / Legal

Order No. 171 caps petrol and diesel retail prices. Mandatory fuel quotas apply.

E

Economic context

Oil price volatility of $60–$90 per barrel. Domestic inflation control remains important.

S

Social responsibility

Maintaining more than 52,000 jobs across West Kazakhstan and Mangystau.

T

Technological drivers

Need for IoT sensors at ANPZ and PNHZ refineries; further development of the ABAI platform.

3 STRATEGIC BUSINESS CONSTRAINTS

No asset divestment

ANPZ, PNHZ and PKOP refineries must remain state-controlled.

No surge pricing

AI-based dynamic retail pricing is prohibited under the stated legal framework.

Labour Code compliance

Workforce reallocation requires union consultation and retraining.

AI role definition

AI serves as a Decision Support System (DSS), with engineer sign-off.

AI could improve operations without changing KMG’s core role

Improve operations while keeping KMG’s core responsibilities and its role in Kazakhstan’s energy supply unchanged.

WHAT COULD CHANGE

1. **Asset maintenance**
   From reactive calendar repairs to machine-learning vibration predictions up to 14 days in advance at ANPZ and PNHZ.
2. **Subsoil reservoir management**
   Scale ABAI digital twin waterflooding across 1,660 injection cells to reduce pump power OPEX.
3. **Industrial safety monitoring**
   Deploy TUMAR computer vision across 60 well workover crews to help prevent HSE incidents.
4. **Supply chain dispatching**
   Automate telemetry ingestion from SCADA directly into SAP ERP maintenance workflows.

WHAT STAYS THE SAME

1. **State ownership structure**
   Samruk-Kazyna (67.42%) and the Ministry of Finance (20%) keep controlling equity and strategic governance.
2. **Domestic supply obligation**
   Refining 21.0 million tonnes of crude a year to supply 80% of Kazakhstan’s domestic fuel needs.
3. **Statutory price compliance**
   Strict adherence to retail price caps under Order No. 171 for AI-92 petrol and diesel.
4. **Workforce protection mandate**
   Preserving more than 52,000 regional jobs across Mangystau through retraining and upskilling.

Some Shell practices could transfer to KMG, while others do not fit its operating context

Verdict	Shell practice	Transferability rationale and evidence for KMG	Operational target / outcome
ADOPT	**Enterprise predictive maintenance and safety vision systems**	Adopt Shell's vibration ML analytics across ANPZ and PNHZ refineries; scale TUMAR computer vision across 60 workover crews.	**−15% unplanned downtime (below 102 hrs/year); zero HSE LTIs.**
ADAPT	**Scope 1&2 decarbonization and hybrid renewable energy**	Adapt Shell's Net-Zero framework to Kazakhstan's coal/gas grid via the 247 MW Zhanaozen solar/wind/gas hybrid plant with Eni and OGMP 2.0 methane abatement.	**98.8% gas utilization; methane emissions reduction.**
PILOT	**Advanced EOR and Slim Hole subsoil technologies**	Pilot digital twin subsoil modeling at mature fields such as Emba and Uzen. Slim Hole drilling is described as cutting CAPEX by 30% (4.6 bn KZT savings).	**30% CAPEX savings in well construction.**
DELAY	**Retail EV charging infrastructure and green hydrogen**	Delay large-scale capital investment until domestic EV adoption matures beyond 5%.	**Avoid unremunerative CAPEX outlay.**
REJECT	**Dynamic fuel surge pricing and refinery asset divestment**	Reject unhedged retail price floating and refinery sales. Order No. 171 mandates price caps; KMG must guarantee 80% domestic fuel supply.	**Prevent social distress and legal violations.**

A four-category model clarifies where people and AI are responsible

Human Only	Human + AI	AI Executes + Human Review	Automated
Exclusive human authority	AI model + executive sign-off	AI diagnostic + human verification	Rules-based automation
• Workforce reallocation in Mangystau • Executive reviews • Labour union negotiations	• Annual CAPEX allocation across subsidiaries • Decarbonization strategy selection	• Refinery pump failure alerts • ABAI waterflooding adjustments across 1,660 cells	• 1 Hz SCADA telemetry collection • Routine environmental compliance reporting

More human control — *More automation*

HUMAN REVIEW POINT (HRP)

A certified Chief Mechanical Engineer physically audits diagnostic telemetry and signs off in SAP ERP before any refinery processing unit is shut down.

ESCALATION PATH PROTOCOL

15 min — Senior Maintenance Technician

→

1 hour — Chief Mechanical Engineer

→

4 hours — General Director (if repair time exceeds 24 h)

MANUAL FALLBACK PROTOCOL

If IoT networks or AI servers fail (cyber or power outage), operators revert to hourly manual logbooks and analog gauge walks.

Automated sensor data could reduce the 12-24 hour delay in manual analysis

CURRENT PROCESS AT ANPZ (REACTIVE)

12–24 h lag

1 Manual operator walks

Technicians manually record physical pressure and vibration gauges.

2 Excel transcription delay

Data is transcribed into spreadsheets with a 12-24 hour delay.

3 Reactive maintenance

Equipment failure occurs before manual trend analysis detects wear.

4 Unplanned downtime

Average baseline of 120 hours per year of emergency refinery outages.

PROPOSED PROCESS WITH AI SUPPORT (PREDICTIVE)

No manual lag

1 1 Hz IoT telemetry

Continuous automated sensor streaming from refinery pumps.

2 ABAI ML anomaly detection

Machine learning identifies micro-friction up to 14 days before failure.

3 Human Review Point

Chief Mechanical Engineer verifies the anomaly flag within 15 minutes.

4 SAP ERP work order

Scheduled preventive repair in a planned maintenance window.

PROPOSED TARGETS

−15% unplanned refinery downtime

< 102 h/yr target downtime (baseline 120 h/yr)

−10% maintenance OPEX by 2027

Customer-facing digital tools must work within the existing fuel price rules

CUSTOMER JOURNEY IMPROVEMENTS · QAZAQ OIL

Qazaq Oil mobile app

- Digital loyalty program
- Automated contactless payment
- Real-time station fuel availability maps

Fleet management B2B portal

- Automated telemetry fuel card limits
- Consumption analytics
- Automated invoice dispatch for fleets

AI inventory dispatching

- Predictive tank replenishment modeling
- Fewer out-of-stock events in peak harvest seasons

EV charging pilot readiness

- Monitor regional EV adoption
- Hybrid charging hubs on high-density highway corridors when commercially viable

CUSTOMER PROTECTION AND LEGAL BOUNDARIES

No AI-based surge pricing

- KMG should not copy Shell's dynamic pricing approach.
- Order No. 171 sets statutory price caps for AI-92 and diesel.

Agricultural fuel price protection

- Guaranteed preferential diesel allocation for regional farmers during spring sowing and autumn harvest.

Consumer data privacy

- Customer loyalty data encrypted locally
- No monetization or third-party sharing of customer location data

Equal access guarantee

- Uninterrupted fuel supply across remote rural regions regardless of station commercial margin.

Training and redeployment would help employees work with new digital tools

1 Human skills
Socio-political and leadership

- Ethical decision-making
- Union negotiation
- Safety leadership
- Regional stakeholder empathy

2 Business skills
Commercial and operational

- Capital allocation
- Supply chain bottleneck analysis
- PESTEL risk evaluation
- P&L management

3 Digital skills
Technical and ML proficiency

- IoT SCADA telemetry interpretation
- ABAI platform navigation
- Predictive maintenance sign-off

4 Meta skills
Adaptive and problem solving

- Human–AI collaboration
- Algorithmic bias detection
- Manual fallback execution
- Continuous learning

HIRING, TRAINING AND REDEPLOYMENT PLAN

MONTHS 0–3 · Selective hiring

- Recruit 15 specialized Data Engineers and Cyber Reliability Experts
- Establish a central KMG AI Governance Unit

MONTHS 4–8 · Enterprise retraining

- Mandatory 40-hour ML diagnostic certification for 150 refinery engineers at ANPZ
- Train 200 reservoir engineers on the ABAI platform

MONTHS 9–12 · Internal redeployment

- Redeploy administrative staff whose work is reduced by SCADA automation into HSE compliance and community liaison roles
- Aim to avoid compulsory redundancies

The main controls address cyber risk, model errors and operational continuity

Risk dimension	Potential operational impact	KMG mitigating safeguard and governance control
1 Privacy risk	Exposure of employee or customer PII records.	Role-based access controls, VPN encryption, local Astana Data Center hosting.
2 Cybersecurity risk	Ransomware attack on refinery OT networks.	Air-gapped SCADA control systems, 2FA, daily offsite encrypted backups.
3 Model bias risk	ML trained on summer telemetry misinterpreting winter wear.	Weekly model recalibration using 5-year seasonal climate datasets (−30°C).
4 Reliability risk	False-positive shutdown alert or missed bearing wear.	Mandatory Human Review Point requiring physical vibrometer sign-off.
5 Accountability gap	Engineers deflecting blame onto AI recommendations.	Explicit policy designating AI as a Decision Support System requiring engineer signature.
6 Escalation failure	System failing to alert control room during rapid vibration.	Dual redundant alarm circuits on independent physical networks.
7 Fallback risk	Operator skill degradation due to over-automation.	Mandatory monthly manual control room drills and analog gauge walks.

The estimated business case depends on 375 million KZT in investment and projected operating savings

TOTAL TRANSFORMATION CAPEX

375 m KZT

ESTIMATE

≈ $720,000 (approximate)

NET PRESENT VALUE ESTIMATE

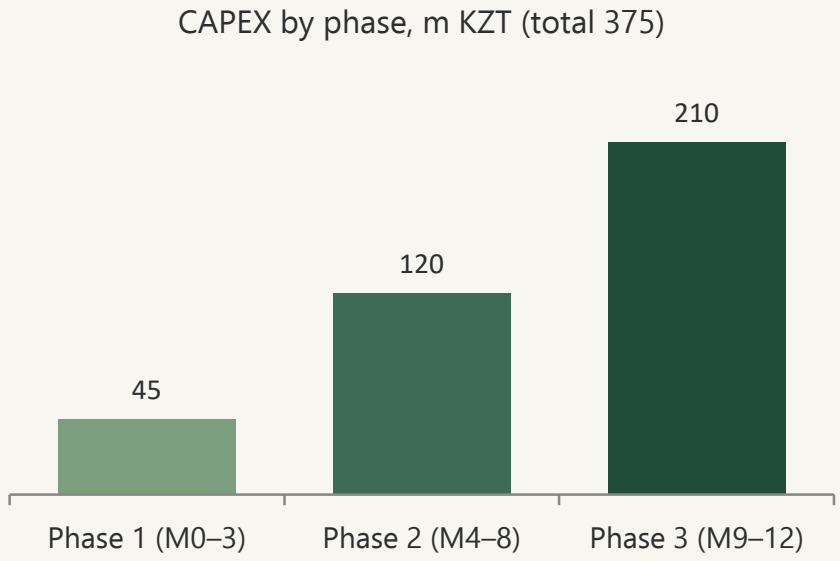
1.42 bn KZT

3-year horizon; 8–12% reservoir OPEX cut plus outage savings

CAPITAL PAYBACK PERIOD ESTIMATE

7.4 months

Linked to avoided emergency downtime of 500 m KZT per day



Avoided outage benefit

ANPZ emergency downtime costs about 500 m KZT per day in lost throughput. A 15% reduction (below 102 h/yr) is estimated to save about 375 m KZT a year.

Artificial-lift energy benefit

Scaling ABAI across 1,660 injection cells is expected to cut pump electricity use by 8–12%. Estimated annual savings: about 280 m KZT.

Balance sheet protection

The 375 m KZT CAPEX is funded from 1,848 bn KZT of liquid cash reserves. Net Debt / EBITDA is expected to stay at about 0.16x.

IRR and discount rate

Estimated IRR: 42%. WACC: 10%. Evaluation period: 36 months.

The KPI dashboard tracks operational, financial and safety results

Metric	Baseline	Target	Responsible owner	Measurement	Strategic area
Unplanned refinery downtime	120 hrs/year	**Below 102 hrs/year (−15%)**	Chief Mechanical Engineer	Quarterly	ANPZ / PNHZ operations
SCADA telemetry data quality	82%	**Above 98% clean data**	Chief Information Officer	Monthly	Data infrastructure
ABAI waterflooding expansion	0 cells	**1,660 cells deployed**	Head of Subsoil Operations	Bi-weekly	Upstream production
Refinery maintenance OPEX	Current annual maintenance expenditure to be established using historical financial records.	**−10% reduction**	Chief Financial Officer	Annual	Financial efficiency
Workover safety LTIs (TUMAR)	0 lost time incidents	**0 lost time incidents**	Chief HSE Officer	Continuous	Industrial safety

UNPLANNED DOWNTIME: BASELINE TO TARGET (HOURS PER YEAR)



The 12-month plan moves from data preparation to a controlled pilot and then wider rollout

PHASE 1 · FOUNDATION

Months 0–3

ACTION Audit and clean SCADA sensor telemetry across ANPZ.
OWNER Chief Information Officer; Lead Data Engineer.
KPI Data quality score from 82% to above 98% by Month 3.
DEPENDS ON Access to SAP ERP historical maintenance records.
COST 45 m KZT for data auditing software and local staff.
RISK / CONTROL Incompatible legacy sensors; custom API translation.

GATE 1: proceed only if data quality score exceeds 95%.

PHASE 2 · PILOT DEPLOYMENT

Months 4–8

ACTION Pilot ABAI predictive failure ML at ANPZ catalytic cracking.
OWNER Chief Mechanical Engineer; Lead Data Scientist.
KPI 14-day prediction lead time; 10% downtime reduction by Month 8.
DEPENDS ON Successful Phase 1 data audit sign-off.
COST 120 m KZT for IoT sensors and ABAI integration.
RISK / CONTROL False alarms; mandatory physical vibrometer check.

GATE 2: scale only if false-positive rate is below 5%.

PHASE 3 · ENTERPRISE SCALING

Months 9–12

ACTION Scale predictive failure AI across PNHZ and 1,660 ABAI cells.
OWNER VP Refining Operations; KMG Engineering Lead.
KPI −15% refinery downtime (below 102 h/yr); −10% OPEX.
DEPENDS ON Phase 2 pilot sign-off by Chief Engineer.
COST 210 m KZT for software scaling and staff training.
RISK / CONTROL Model drift at −30°C; five-year seasonal dataset and recalibration.

GATE 3: maintain programme if annual OPEX savings exceed 8%.

The proposal depends on coordination between operations, IT, employees and risk teams

1 Technical operations and IT

SCADA IoT ingestion
Feeds continuous 1 Hz vibration data into the ABAI ML engine.

SAP ERP integration
Automates work order logging after Chief Engineer sign-off.

2 Workforce and competency

Engineer certification
150 refinery engineers certified on ML diagnostic sign-off.

Union liaison
Mangystau labour unions engaged in zero-layoff redeployment.

Predictive maintenance programme

ANPZ · PNHZ · ABAI

3 Governance and risk control

Air-gapped security
Complete physical isolation of SCADA OT networks from the public web.

HRP audit logs
Immutable digital logs of every engineer diagnostic sign-off.

4 Business case and roadmap

Decision gates 1–3
Financial release tied to audit data quality score above 95%.

OPEX tracking
Real-time auditing of energy savings across 1,660 ABAI cells.

AI-generated claims were checked against company and regulatory sources before use

Topic / metric	AI initial output	Human verification and audit correction	Source citation
Debt-to-Equity ratio	AI reported 0.92x D/E and warned of imminent insolvency risk.	Human audit found AI aggregated joint venture debt. Stated verified D/E is 0.24x.	*KMG audited financials (PwC 2025)*
Net Debt / EBITDA	AI reported Net Debt / EBITDA of 0.45x.	Human audit corrected the cash deduction using 1,848 bn KZT in cash. Corrected ratio is 0.16x.	*KMG Annual Report 2025*
Retail price strategy	AI suggested dynamic peak-demand surge pricing at Qazaq Oil stations.	Human audit rejected the claim as illegal under the Order No. 171 price caps.	*Minister of Energy Order No. 171*
Refinery outage downtime	AI claimed ANPZ suffers 400 hours per year of downtime.	Human audit states the baseline is 120 hours per year; target is below 102.	*ANPZ operational technical logs 2025*

Key message: AI outputs are decision-support inputs, not verified facts. Human validation is essential for financial, legal and operational decisions.

The project should be tested against changes in oil prices, winter conditions and cyber incidents

SCENARIO A

Low oil price ($50/bbl)

TESTED CONDITION

Crude falls to $50 per barrel; KMG EBITDA falls by 25%.

BUSINESS IMPACT

CAPEX budgets are tightened across regional subsidiaries.

KMG MITIGATION

Protect the 375 m KZT AI CAPEX. ABAI waterflooding reduces OPEX by 12%, helping preserve cash.

VERDICT

Project remains NPV-positive at 1.1 bn KZT (base case: 1.42 bn KZT).

SCENARIO B

Sub-zero winter shock (−35°C)

TESTED CONDITION

Extreme freeze at ANPZ causes sensor drift.

BUSINESS IMPACT

Possible false-positive vibration alerts in pump ML models.

KMG MITIGATION

Five-year winter datasets, regular recalibration, physical vibrometer checks and engineer sign-off.

VERDICT

Engineer review is meant to stop shutdowns based only on unverified alerts.

SCENARIO C

Cyber network interruption

TESTED CONDITION

Ransomware attack or OT network communication outage.

BUSINESS IMPACT

Loss of real-time IoT telemetry feeds to the ABAI engine.

KMG MITIGATION

Trigger the Manual Fallback Protocol at once: hourly operator logbooks and analog gauge walks.

VERDICT

Manual procedures are meant to keep operators in control while digital systems are restored.

References and project information

ACADEMIC AND CORPORATE REFERENCES (IEEE)

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[2] JSC NC KazMunayGas, “Annual Integrated Report 2025: Operational and Sustainability Performance,” Astana, 2025.

[3] Shell plc, “Annual Report and Accounts 2025: Strategic Progress and Financial Statements,” London, UK, 2025.

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DOCUMENT CONTROL AND DECLARATION

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